FROZEN BLITZ · TIDECHALLENGE 2026

Team Update 01

Thursday, November 5, 2026

GENERAL

We forgot AprilTags. They are now on the FIELD: 1 on each outward-facing wall of the SCOOPING STATIONS and the DELIVERY ZONE structures, 8 in total. They use the same family and size as the 2026 FIRST Robotics Competition FIELD, so your vision code only needs the new layout.

The END GAME extension exception now starts as soon as a ROBOT is partially within its CHECKOUT ZONE or touching its CHECKOUT ZONE tables, so ROBOTS can extend to climb onto the tables.

> Additions are highlighted in yellow. Deletions are indicated with a ~~strikethrough~~. Notes that explain the change but are not part of the manual are in **bold**.

The following documents were updated with this Team Update:

Game Manual v1.5 PDF and web version	**FIELD drawings, Rev 1.5** new sheet FB-13, AprilTag layout
FIELD CAD model AprilTags added on their faces	**Web manual** AprilTags, G418 and R105 updated

SECTION 4 · ARENA

4.2 FIELD

The FIELD is populated with the following elements:

- 2 FREEZERS (a red FREEZER and a blue FREEZER),
- 2 STORAGES (a red STORAGE and a blue STORAGE),
- 2 DELIVERY ZONES (a red DELIVERY ZONE and a blue DELIVERY ZONE),
- 2 SCOOPING ZONES (a red SCOOPING ZONE and a blue SCOOPING ZONE), each with a SCOOPING STATION, ~~and~~
- 2 CHECKOUT ZONES (a red CHECKOUT ZONE and a blue CHECKOUT ZONE), divided by the CENTER LINE~~.~~, and
- 8 AprilTags (see section 4.9 AprilTags).

4.9 AprilTags

New section.

AprilTags help ROBOTS find their position on the FIELD. FROZEN BLITZ uses the same AprilTag family and size as the 2026 FIRST Robotics Competition FIELD, so existing vision code only needs the FROZEN BLITZ FIELD layout.

- All markers are from the 36h11 tag family, IDs 1 to 8. There are 8 unique markers on the FIELD, and no other IDs are used.

- Each AprilTag is an 8.125in (~20.64cm) square target, including its white border; the black square is 6.5in (~16.51cm).
- AprilTags are printed on a flat, matte material and mounted vertically, centered on the face they are on, facing into the FIELD.
- 4 AprilTags are on the SCOOPING STATIONS, 1 on each of the 2 outward-facing walls of each SCOOPING STATION. Their top edge is 1.0in (~2.5cm) below the top of the wall.
- 4 AprilTags are on the DELIVERY ZONE structures, 1 on each of the 2 outward-facing angled walls of each structure. Their top edge is 1.0in (~2.5cm) below the top of the wall.

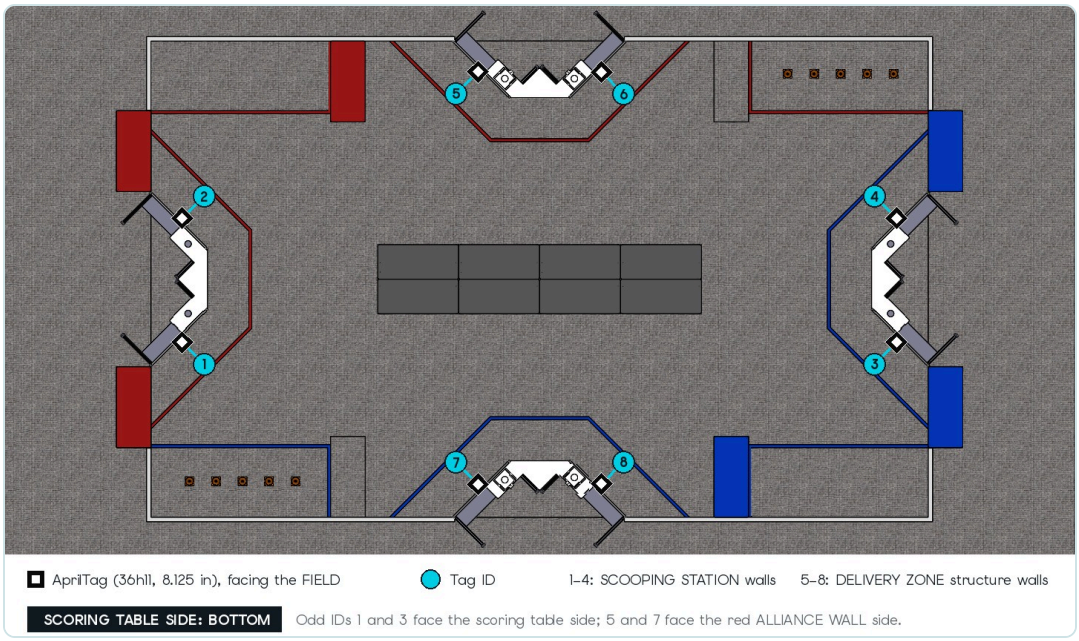


Figure 4-16 *AprilTag locations*

Table 4-2 *AprilTag IDs*

ID	FIELD element	Face
1	red SCOOPING STATION	wall facing the scoring table side
2	red SCOOPING STATION	wall facing the far side
3	blue SCOOPING STATION	wall facing the scoring table side
4	blue SCOOPING STATION	wall facing the far side
5	red DELIVERY ZONE structure	wall on the red ALLIANCE WALL side
6	red DELIVERY ZONE structure	wall on the blue ALLIANCE WALL side
7	blue DELIVERY ZONE structure	wall on the red ALLIANCE WALL side
8	blue DELIVERY ZONE structure	wall on the blue ALLIANCE WALL side

The exact position of every AprilTag is in the official CAD model and FIELD drawings. If AprilTags wear or get marked during MATCHES, they are repaired with gaffers tape.

> ROBOTS staged against a SCOOPING STATION wall (per G303) and ROBOTS working at a DELIVERY ZONE may cover AprilTags. Plan your vision for it.

SECTION 6 · GAME RULES (G)

G418 Expansion limits.

A ROBOT may not extend beyond the expansion limit described in R105. This rule does not apply during END GAME to a ROBOT that is partially or completely within its ALLIANCE'S CHECKOUT ZONE ~~during END GAME~~, or in contact with its ALLIANCE'S CHECKOUT ZONE tables.

The rest of G418 (exceptions, violation, and Figure 6-2) is unchanged.

SECTION 7 · ROBOT CONSTRUCTION RULES (R)

R105 ROBOT extension limit.

ROBOTS may not extend more than 16.0in (40.64cm) beyond the vertical projection of their ROBOT PERIMETER, except during END GAME while partially or completely within their ALLIANCE'S CHECKOUT ZONE ~~during END GAME~~, or in contact with their ALLIANCE'S CHECKOUT ZONE tables (see G418).

SECTION 10 · GLOSSARY

Term	Definition
APRILTAG	a 36h11 visual marker mounted on the FIELD to help ROBOTS find their position; there are 8 on the FIELD, IDs 1 to 8 (see section 4.9)